THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 2	$2250 = 2 \times 3^2 \times 5^3$
2	**(d)** 1	Each point where the graph meets the x-axis is a zero
3	**(c)** infinitely many solutions (coincident lines)	Compare a_1/a_2, b_1/b_2, c_1/c_2
4	**(b)** 7	Sum $= -b/a$
5	**(b)** -6	$a_n = a + (n-1)d$
6	**(c)** 6	BPT
7	**(b)** 5	Distance formula
8	**(b)** $\frac{4}{5}$	Hypotenuse is 5
9	**(c)** $15\sqrt{3}$ m	$h = 15\tan 60^\circ = 15 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 15\sqrt{3}$ m
10	**(d)** 50°	$\angle AOB = 100^\circ$ and OP bisects it
11	**(a)** ₹ 880	Cost $= 88 \times 10$; Cost = ₹ 880
12	**(a)** 858 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 7 \times 25 + 2 \times \frac{22}{7} \times 7^2 = 858$ cm²
13	**(d)** 44.2	$\sum f_i = 50$ and $\sum f_i x_i = 2210$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{2210}{50} = 44.2$
14	**(a)** $\frac{5}{36}$	Favourable outcomes: (2,6), (3,5), (4,4), (5,3), (6,2); number $= 5$; $P = \frac{5}{36}$
15	**(b)** 22338	LCM $= \dfrac{306 \times 657}{9}$
16	**(d)** 5	Sum of zeroes $= 7$, so other $= 7 - 2 = 5$
17	**(b)** 5	Solve by elimination
18	**(c)** 9	$D = 0$: $4m^2 - 4k = 0$
19	**(b)** 84	Reverse AP has difference $-d$
20	**(b)** 9	Sides in ratio $\sqrt{\text{areas}}$

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$6 = 2 \times 3$ has no factor 5 (1); so 6^n never ends in 0 (1).	1+1
22	Zeroes are -2 and 3 (1); a parabola, so degree 2 (1).	1+1
23	Let father x, daughter y now (1); $x - 7 = 7(y - 7)$ and $x + 3 = 3(y + 3)$ (1).	1+1
24	Integers $x, x+1$: $x(x+1) = 72$ (1); $x^2 + x - 72 = 0$ (1).	1+1
25	$a_n = 3 + (n-1)(5)$ (1); $n = 13$ (1).	1+1
26	$d^2 = 289 - 225 = 64$ (1); $d = 8$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Assume $\sqrt{2} = a/b$ in lowest terms, $b \neq 0$ (1); then $2b^2 = a^2$, so $2 \mid a^2$ hence $2 \mid a$ (1); put $a = 2c$: $b^2 = 2c^2$ so $2 \mid b$ - contradicts coprime (1).	1+1+1
28	Zeroes -2, 3 (1); sum $1 = -b/a$ (1); product $-6 = c/a$ (1).	1+1+1
29	$\frac{a_1}{a_2} = \frac{5}{7}$, $\frac{b_1}{b_2} = -\frac{2}{3}$, $\frac{c_1}{c_2} = \frac{8}{9}$ (2); ratios $a_1/a_2 \neq b_1/b_2$: lines intersect (1).	2+1
30	$D = (-7)^2 - 4(2)(3) = 25$ (1); $x = \dfrac{7 \pm 5}{4}$ (1); roots $x = 3, \frac{1}{2}$ (1).	1+1+1
31	$a + d = 7$, $a + 4d = 19$ (2); subtracting, $d = 4$, $a = 3$ (2); $a_{10} = 39$ (1).	2+2+1
	OR: By the Basic Proportionality Theorem, $\dfrac{AD}{DB} = \dfrac{AE}{EC}$ (1); $\dfrac{4}{6} = \dfrac{12}{EC}$ (1); $EC = 18$ cm (1).	1+1+1
32	Area of triangle must be 0 (1); $\frac{1}{2}\lvert 1(6-k) + 3(k-2) + 5(2-6)\rvert = 0$ (1); $k = 10$ (1).	1+1+1

	OR: Hypotenuse $= \sqrt{12^2+5^2} = 13$ (1); $\sin\theta = \frac{12}{13}$, $\cos\theta = \frac{5}{13}$ (1); $\sec\theta = \frac{13}{5}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Assume $\sqrt{17} = a/b$, with a, b coprime integers, $b \neq 0$ (1); squaring, $17b^2 = a^2$, so $17 \mid a^2$ and hence $17 \mid a$ (1); write $a = 17c$: $b^2 = 17c^2$, so $17 \mid b$ (1); then 17 is a common factor of a and b, contradicting coprimality (1); so $\sqrt{17}$ is irrational (1).	1+1+1+1+1
34	$\alpha+\beta = 7$, $\alpha\beta = 10$ (1+1); new sum $= 9$ (1); new product $= \alpha\beta + (\alpha+\beta) + 1 = 18$ (1); polynomial $x^2 - (9)x + (18)$ (1).	1+1+1+1+1
35	Son x, father $8x$ (1); $8x + 9 = 5(x+9)$ (2); $x = 12$ (1); son 12 years, father 96 years (1).	1+2+1+1
36	Speed x: $\dfrac{360}{x} - \dfrac{360}{x+5} = 1$ (1); $1800 = x(x+5)$ so $x^2 + 5x - 1800 = 0$ (2); $(x-40)(x+45) = 0$ (1); speed 40 km/h (1).	1+2+1+1
	OR: $S_n = \frac{n}{2}[2(24) + (n-1)(-2)] = 144$ (1); form quadratic in n (2); $n = 9$ (the other root is rejected) (2).	1+2+2
37	Given $\triangle ABC$, $DE \parallel BC$; join BE, CD and draw $EM \perp AB$, $DN \perp AC$ (1); $\frac{ar(ADE)}{ar(BDE)} = \frac{\frac{1}{2}AD\cdot EM}{\frac{1}{2}DB\cdot EM} = \frac{AD}{DB}$ and $\frac{ar(ADE)}{ar(CDE)} = \frac{AE}{EC}$ (2); $ar(BDE) = ar(CDE)$ (same base DE, same parallels) (1); hence $\frac{AD}{DB} = \frac{AE}{EC}$ (1).	1+2+1+1
	OR: Let the point be $(x, 0)$; $(x-7)^2 + 36 = (x+3)^2 + 16$ (2); $x = 3$ (2); point $(3, 0)$ (1).	2+2+1
38	LHS $= \dfrac{\sin^2\theta + (1+\cos\theta)^2}{\sin\theta(1+\cos\theta)}$ (1); $= \dfrac{\sin^2\theta + 1 + 2\cos\theta + \cos^2\theta}{\sin\theta(1+\cos\theta)} = \dfrac{2+2\cos\theta}{\sin\theta(1+\cos\theta)}$ (2); $= \dfrac{2}{\sin\theta} = 2\csc\theta$ = RHS (2).	5
	OR: Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 80 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 80$ gives $h = 20\sqrt{3}$ (2); Height $20\sqrt{3}$ m; distances 20 m and 60 m (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** $(2, 3)$	Section formula
2	**(a)** $\frac{12}{5}$	Find the perpendicular
3	**(a)** $3\sqrt{3}$ m	Height $= 6\sin 60^\circ = 6 \times \frac{\sqrt{3}}{2}$; Height reached on the wall $= 3\sqrt{3}$ m
4	**(a)** 30 cm	$PT^2 = OP^2 - OT^2 = 34^2 - 16^2 = 900$; $PT = 30$ cm
5	**(b)** 50	Distance 66 m $= 6600$ cm; revolutions $= \frac{6600}{132}$; Revolutions $= 50$
6	**(d)** 374 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 3.5 \times 10 + 4 \times \frac{22}{7} \times 3.5^2$; TSA $= 374$ cm^2
7	**(b)** 78.75	Median class 70–80: $l = 70$, $cf = 9$, $f = 4$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 70 + \frac{12.5 - 9}{4} \times 10 = 78.75$
8	**(d)** $\frac{3}{8}$	Favourable outcomes: HHT, HTH, THH; number $= 3$; $P = \frac{3}{8}$
9	**(b)** always irrational	If it were rational, dividing by the rational would make the irrational rational
10	**(a)** $x^2 - 7x + 12$	$x^2 - (\text{sum})x + \text{product}$
11	**(d)** 2	Solve by elimination
12	**(b)** 14	Other root 7; product $= k$
13	**(c)** 3	$a_{n+1} - a_n = 3$
14	**(d)** 18	$d^2 = s^2 + s^2$
15	**(c)** 12	$\frac{1}{2} \times$ base $\times$ height
16	**(b)** 1	Substitute the standard values
17	**(c)** 12 m	Add the height of the eyes: $2 + 10$; Height of building $= 12$ m
18	**(a)** 5 cm	$9 + 7 = AD + 11$; $AD = 5$ cm
19	**(a)** 28 cm	$R = r_1 + r_2 = 19 + 9$; $R = 28$ cm
20	**(c)** 1330 cm²	$6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$ with $r = 7$ cm; TSA $= 1176 + 154 = 1330$ cm^2

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (1); $= \sqrt{(2)^2 + (-2)^2} = \sqrt{8}$ (1); $= \sqrt{8}$ units (1).	1+1+1
22	Hypotenuse $= 5$ (1); $\sec A = \frac{5}{3}$ (1).	1+1
23	Broken part (hypotenuse) $= \frac{6}{\sin 30^\circ} = 12$ m; Distance from the foot $= 6\cot 30^\circ = 6\sqrt{3}$ m (1); Distance from the foot of the tree to the point where its top touches the ground $= 6\sqrt{3}$ m (1).	1+1
24	Tangents from an external point are equal: $QA = QB$; $3x + 4 = 2x + 9$ gives $x = 5$ (1); $QA = 3(5) + 4 = 19$ cm (1).	1+1
25	Angle turned in 15 minutes $= 90^\circ$; Arc $= \frac{90}{360} \times 2 \times \frac{22}{7} \times 14$ (1); Distance $= 22$ cm (1).	1+1
26	Volume of sphere $=$ volume of wire (cylinder) $= \pi r^2 L$; $\frac{4}{3}\pi(6)^3 = \pi(2)^2 L$ (1); $L = 72$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 20 m (1); $h(\cot 30^\circ - \cot 60^\circ) = 20$ gives $h = 20 \div (\frac{2\sqrt{3}}{3})$ (1); Height $= 10\sqrt{3}$ m (distances 30 m and 10 m) (1).	1+1+1
28	Tangent lengths from a vertex are equal: $AF = AE = x$, $BD = BF = y$, $CD = CE = z$ (1); $x + y = 12$, $y + z = 7$, $z + x = 9$; adding, $x + y + z = 14$ (1); $AF = 7$ cm, $BD = 5$ cm, $CE = 2$ cm (1).	1+1+1
29	Segment $=$ sector $-$ triangle (1); Sector $= 38.5$ cm^2, triangle $= \frac{1}{2} \times 7 \times 7 = 24.5$ cm^2 (1); Segment $= 38.5 - 24.5 = 14$ cm^2 (1).	1+1+1
30	Volume $=$ hemisphere $+$ cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 19404$ cm^3; cone $= \frac{1}{3}\pi r^2 h = 4620$ cm^3 (1); Volume $= 24024$ cm^3 (1).	1+1+1
31	Take assumed mean $a = 35$; deviations $d_i = x_i - a$: -30, -20, -10, 0, 10, 20 (1); $\sum f_i = 100$ and $\sum f_i d_i = -880$ (1); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 35 + \frac{-880}{100} = 26.2$ (1).	1+1+1

	OR: Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (1,1), (1,4), (2,2), (3,3), (4,1), (4,4), (5,5), (6,6); number $= 8$ (1); $P = \frac{8}{36} = \frac{2}{9}$ (1).	1+1+1
32	Assume $3 + 2\sqrt{5} = r$ rational (1); then $\sqrt{5} = \dfrac{r-3}{2}$ is rational (1) - contradiction; so it is irrational (1).	1+1+1
	OR: $p(2) = 0$: $4 - 10 + 6 = 0$ gives $k = -5$ (1); sum of zeroes $= -k = 5$ (1); other zero $= 5 - (2) = 3$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Let the tangent XY touch the circle with centre O at P, and take any other point Q on XY (1); Q lies outside the circle (a tangent meets the circle only at P), so $OQ > OP$ (1); This holds for every point $Q \neq P$ on XY, so OP is the shortest distance from O to the line XY (2); The shortest distance is the perpendicular, so $OP \perp XY$ (1).	1+1+2+1
34	Segment $=$ sector $-$ triangle (2); Sector $= 102.67$ cm^2; triangle $= \frac{\sqrt{3}}{4}r^2 = 84.77$ cm^2 (2); Segment $= 102.67 - 84.77 = 17.90$ cm^2 (1).	2+2+1
35	Volume of water is the same in both tanks (2); $\pi r^2 h = 1540$ cm^3 and base area of cuboid $= 77$ cm^2 (2); Height $= 1540 \div 77 = 20$ cm (1).	2+2+1
36	$\sum f_i = 21 + p$ and $\sum f_i x_i = 1535 + 75p$ (2); $\frac{1535+75p}{21+p} = 73.4$ gives $(75 - 73.4)p = 6.4$ (2); $p = 4$ (1).	2+2+1
	OR: Total outcomes $= 36$ (2); Favourable outcomes: 5; 6; 4 (2); Probabilities: $\frac{5}{36}$, $\frac{1}{6}$, $\frac{1}{9}$ (1).	2+2+1
37	Maximum weight $=$ HCF$(420, 130) = 10$ kg (3); trays $= \dfrac{420}{10} + \dfrac{130}{10} = 42 + 13 = 55$ (2).	3+2
	OR: Sum of zeroes $= -b/a = 8$ (2); third zero $= 8 - (1) - (2) = 5$ (2); check by product $= 10$ (1).	2+2+1
38	Fixed x, per km y: $x + 10y = 170$, $x + 15y = 240$ (2); subtract: $5y = 70$, $y = 14$ (1); $x = 30$ (1); fare for 25 km $= 30 + 14 \times 25 = 380$ (1).	2+1+1+1
	OR: Breadth x, length $x + 7$: $x(x + 7) = 330$ (1); $x^2 + 7x - 330 = 0$ (1); $(x - 15)(x + 22) = 0$ (1); $x = 15$ m (breadth), length 22 m (reject negative) (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 72.5	$l=70,\ f_1=13,\ f_0=10,\ f_2=4,\ h=10$; Mode $= l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h = 70+\frac{3}{12}\times 10 = 72.5$
2	**(b)** $\frac{4}{11}$	Favourable letters: A, E, A, I; number $=4$; $P=\frac{4}{11}$
3	**(c)** 5	$3125=5^5$; the number of places is the larger of the powers of 2 and 5
4	**(b)** 9	$\alpha+\beta=-b/a$
5	**(c)** 24	$x=6,\ y=4$
6	**(d)** 3 and 5	Factorise
7	**(c)** 3	$a_1=S_1$
8	**(d)** 25	Pythagoras
9	**(a)** 8	$x=2m-a$
10	**(b)** 1	$\tan(90^\circ-\theta)=\cot\theta$
11	**(d)** $10\sqrt{3}$ m	$h(\cot 30^\circ-\cot 60^\circ)=20$ gives $h=20\div(\frac{2\sqrt{3}}{3})$; Height $=10\sqrt{3}$ m (distances 30 m and 10 m)
12	**(b)** 30°	$\angle OAP=90^\circ$ (radius $\perp$ tangent); $\angle OAB=90^\circ-60^\circ=30^\circ$
13	**(c)** 10 cm	$R^2=6^2+8^2=100$; $R=10$ cm
14	**(b)** 1760 cm²	TSA = CSA cylinder + one flat base + CSA of cone; $=1056+154+550=1760$ cm^2
15	**(c)** 30–40	It belongs to the class 30–40; Modal class = 30–40
16	**(c)** $\frac{1}{2}$	Favourable outcomes = number of red balls; $P=\frac{6}{12}=\frac{1}{2}$
17	**(a)** $\sqrt{5}+3$	Rational + irrational is irrational
18	**(d)** 14	$\alpha\beta=c/a$
19	**(d)** 4	$\frac{1}{2}=\frac{2}{k}\neq\frac{3}{7}$
20	**(b)** 25	$D=b^2-4ac$

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Class marks x_i: 5, 15, 25, 35, 45, 55; $\sum f_i=100$ and $\sum f_ix_i=2620$ (1); Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{2620}{100}=26.2$ (1).	1+1
22	Total outcomes $=2^3=8$: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT; Favourable outcomes: HHT, HTH, HTT, THH, THT, TTH, TTT; number = 7 (1); $P=\frac{7}{8}$ (1).	1+1
23	LCM $=\dfrac{306\times 657}{9}$ (1) $=22338$ (1).	1+1
24	Sum $=-\dfrac{b}{a}=4$ (1); product $=\dfrac{c}{a}=3$ (1).	1+1
25	Express one variable, substitute: $x=5$ (1), $y=3$ (1).	1+1
26	Split the middle term and factorise (1); roots $\frac{3}{2}, 2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Multiply (i) by 4 and (ii) by 3 and subtract: $(-7)x=-21$ (2); $x=3$ (1); put in (i): $y=2$ (1); check in (ii) (1).	2+1+1+1
28	$D=(-6)^2-4(1)(8)=4$ (1); $x=\dfrac{6\pm 2}{2}$ (1); roots $x=4, 2$ (1).	1+1+1
29	$a=18,\ d=-\frac{5}{2},\ l=-\frac{19}{2}$ (1); $l=a+(n-1)d$ gives $n-1=11$ (1); $n=12$ (1).	1+1+1
30	Ratio of areas = (ratio of sides)^2 (1); $\dfrac{16}{36}=\dfrac{4^2}{x^2}$ (1); $x=6$ cm (1).	1+1+1
31	Area $=\frac{1}{2}\lvert x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\rvert$ (1); $=\frac{1}{2}\lvert 3+4-28\rvert$ (1); $=\frac{21}{2}$ sq. units (1).	1+1+1

	OR: $AC = \sqrt{7^2 + 24^2} = 25$ cm (1); $\sin A = \frac{BC}{AC} = \frac{24}{25}$, $\cos A = \frac{AB}{AC} = \frac{7}{25}$ (1); $\tan A = \frac{BC}{AB} = \frac{24}{7}$ (1).	1+1+1
32	Horizontal distance $d = 5\cot 45^\circ = 5$ m (from the depression of the foot) (1); Part of the tower above the top of the building $= d\tan 60^\circ = 5\sqrt{3}$ m (1); Tower $= 5 + 5\sqrt{3}$ m (1).	1+1+1
	OR: $AC = 29$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 20 - r$, $CD = BC - r = 21 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(20 - r) + (21 - r) = 29$ gives $r = 6$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	AP: $a = 100$, $d = 10$, $n = 20$ (1); last $= a + (n-1)d = 290$ (2); total $= S_n = \frac{20}{2}(100 + 290) = 3900$ (2).	1+2+2
34	Ratio of sides = ratio of perimeters $= 1 : 2$ (1); side $= 10$ cm (1); ratio of areas $= 1 : 4$ (1).	1+1+1
35	Diagonals bisect each other: midpoint of AC is $\left(3, \frac{9}{2}\right)$ (2); this is also the midpoint of BD (1); $D = (2, 6)$ (2).	2+1+2
36	Divide numerator and denominator by $\cos A$ (1): $\frac{4\tan A + 3}{4\tan A + 3}$ (1); $= \frac{4(\frac{4}{3}) + 3}{4(\frac{4}{3}) + 3} = 1$ (2); verified with $\sin A = \frac{4}{5}$, $\cos A = \frac{3}{5}$ (1).	1+1+2+1
	OR: Let pedestal $= p$; distance from the point $= p\cot 45^\circ = p$ (2); $p + 2 = p\tan 60^\circ$ gives $p(\sqrt{3} - 1) = 2$ (2); $p = \frac{2}{\sqrt{3}-1} = 1 + \sqrt{3}$ m (1).	2+2+1
37	$AC = \sqrt{6^2 + 8^2} = 10$ cm; $BDOF$ is a square of side r, so $BF = BD = r$ (2); $AF = AE = 6 - r$ and $CD = CE = 8 - r$ and $AE + EC = AC$: $(6 - r) + (8 - r) = 10$ (2); $r = 2$ cm, $AF = 4$ cm, $CD = 6$ cm (1).	2+2+1
	OR: Wire = circumference + the diameters (2); Circumference $= \pi d = 110$ mm; diameters $= 5 \times 35 = 175$ mm (2); Total $= 285$ mm (1).	2+2+1
38	Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(6)^2(8) = \pi(4)^2 x$ (2); $x = 6$ cm (1).	2+2+1
	OR: Median class is 30–40 with $f = 16$, and the cumulative frequency before it is $cf = 18 + x$ (2); $36.25 = 30 + \frac{50-(18+x)}{16} \times 10$ gives $x = 22$ (2); $x = 22$; then $y = 100 - 57 - x = 21$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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